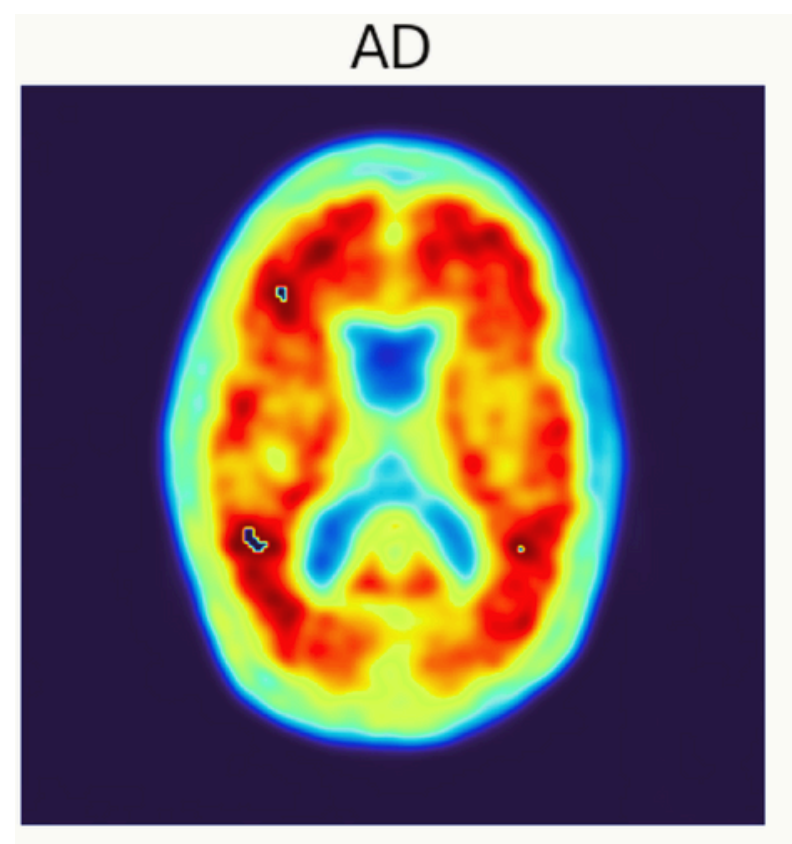




Alzheimer Disease Classification Using PET Images

Comparing DeepLearning Approaches

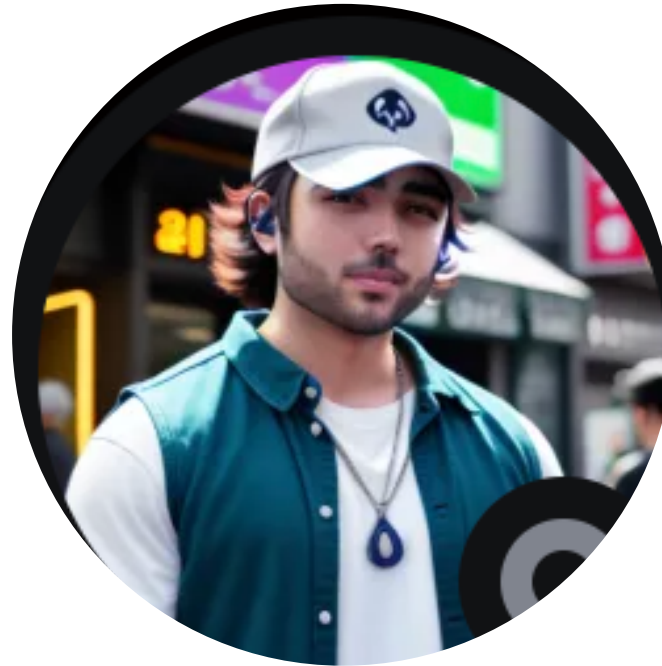


Representative PET Scan of Alzheimer's Disease

Team Members



Abdoulfatah Abdillahi
Member



Russ Mukhtar
Member



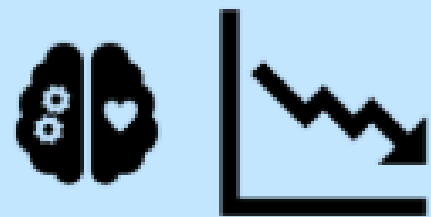
Siham Argaw
Member

Under the supervision of: Dr. Robert Mateescu, Department of Computer Science



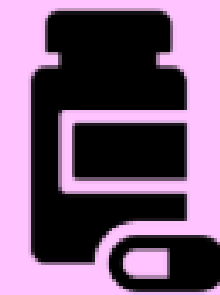
Why should we care about Alzheimer's Disease?

AD is a complex, progressive neurological disease associated with **cognitive, functional, and behavioural** changes



Globally, an estimated **43.8 million people** are living with AD and other dementias costing ~1.3 trillion dollars per year

AD can be a **debilitating disease**, resulting in memory loss and severe interference with activities of daily living



There is **only one disease-modifying therapy for AD** at this time; currently, other treatments are symptomatic



Dataset Overview and Preprocessing

Dataset Source: PET scans from the Alzheimer’s Disease Neuroimaging Initiative (ADNI).



Initial Classes:

- Cognitively Normal (CN),
- Early Mild Cognitive Impairment (EMCI)
- Late Mild Cognitive Impairment (LMCI)
- Alzheimer’s Disease (AD)
- Significant Memory Concerns (SMC)

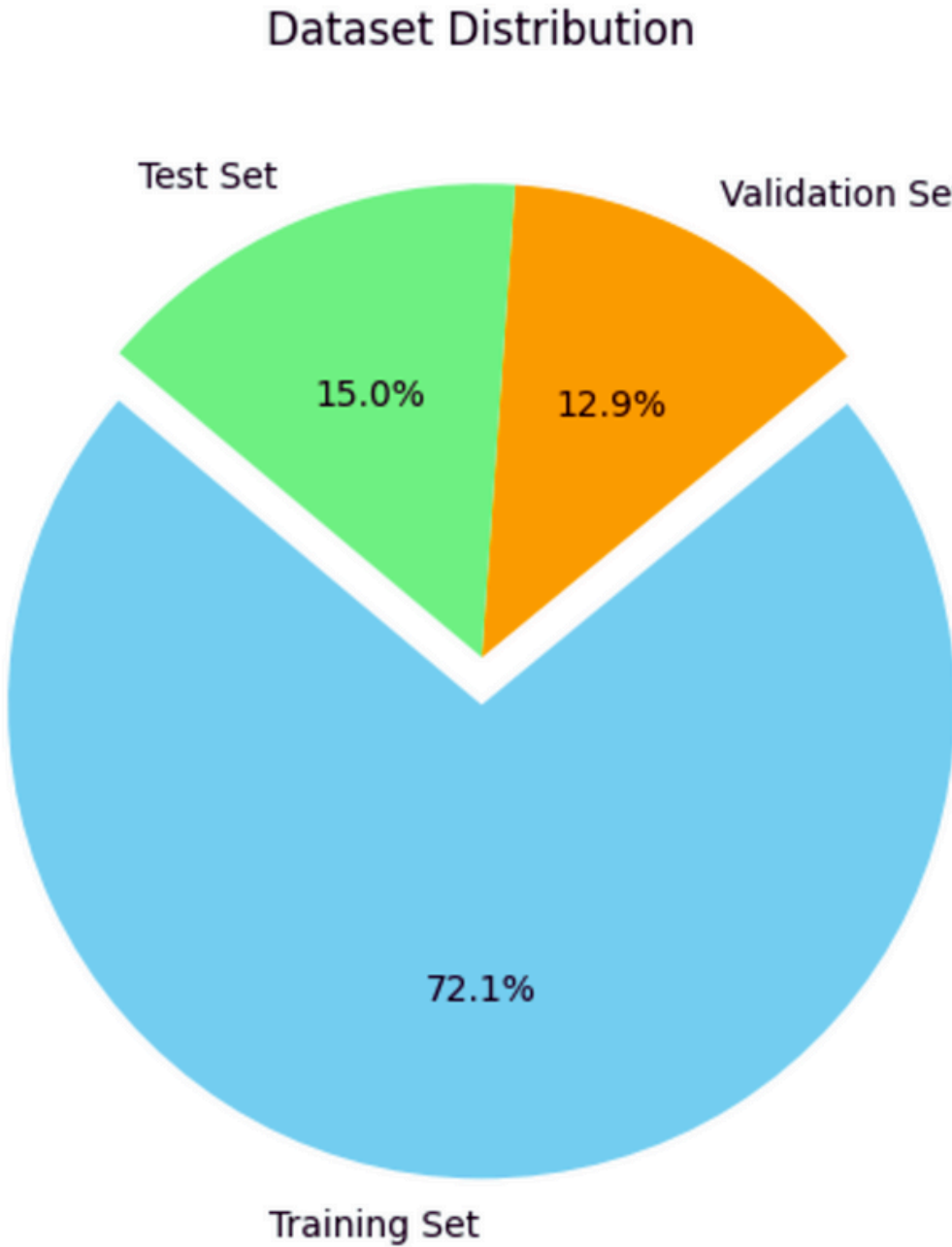
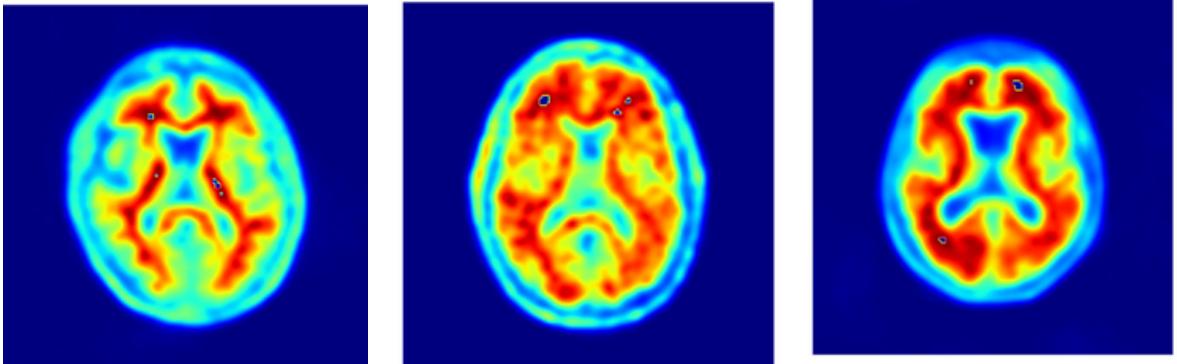
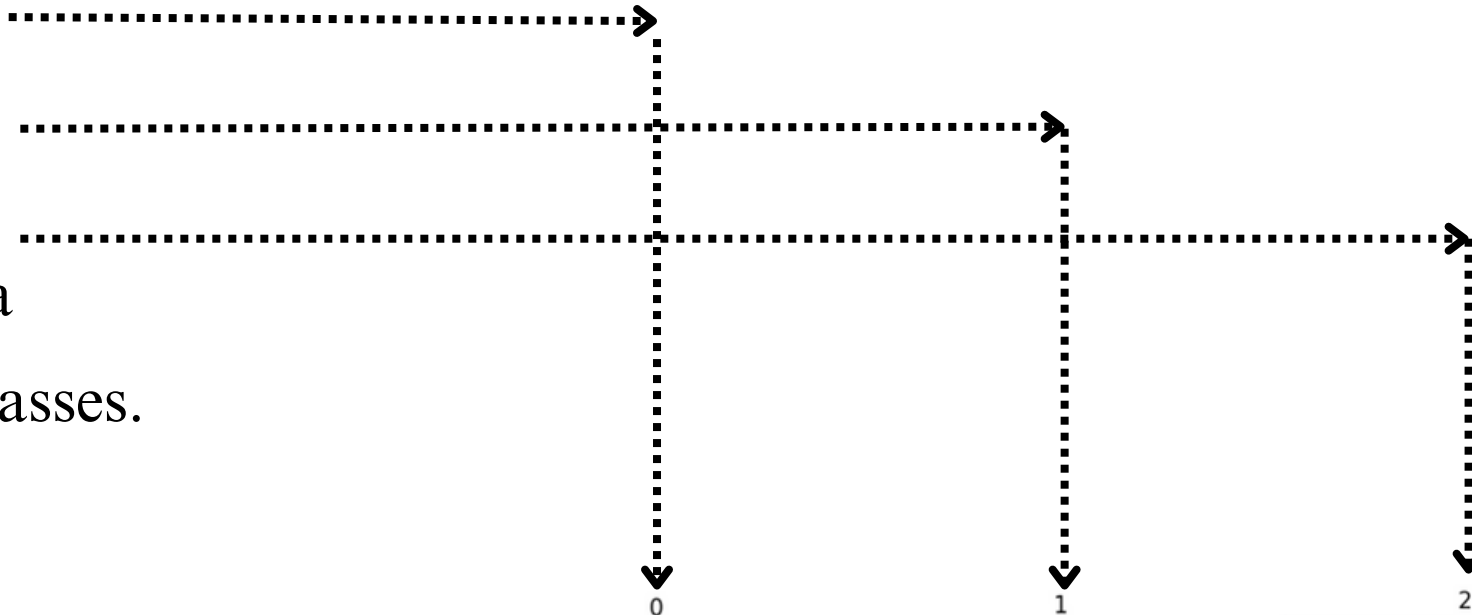
Filtered Classes: Retained three classes:

- CN: 460 patients
- EMCI: 505 patients
- AD: 235 patients

Data Imbalance: Addressed using data augmentation for underrepresented classes.

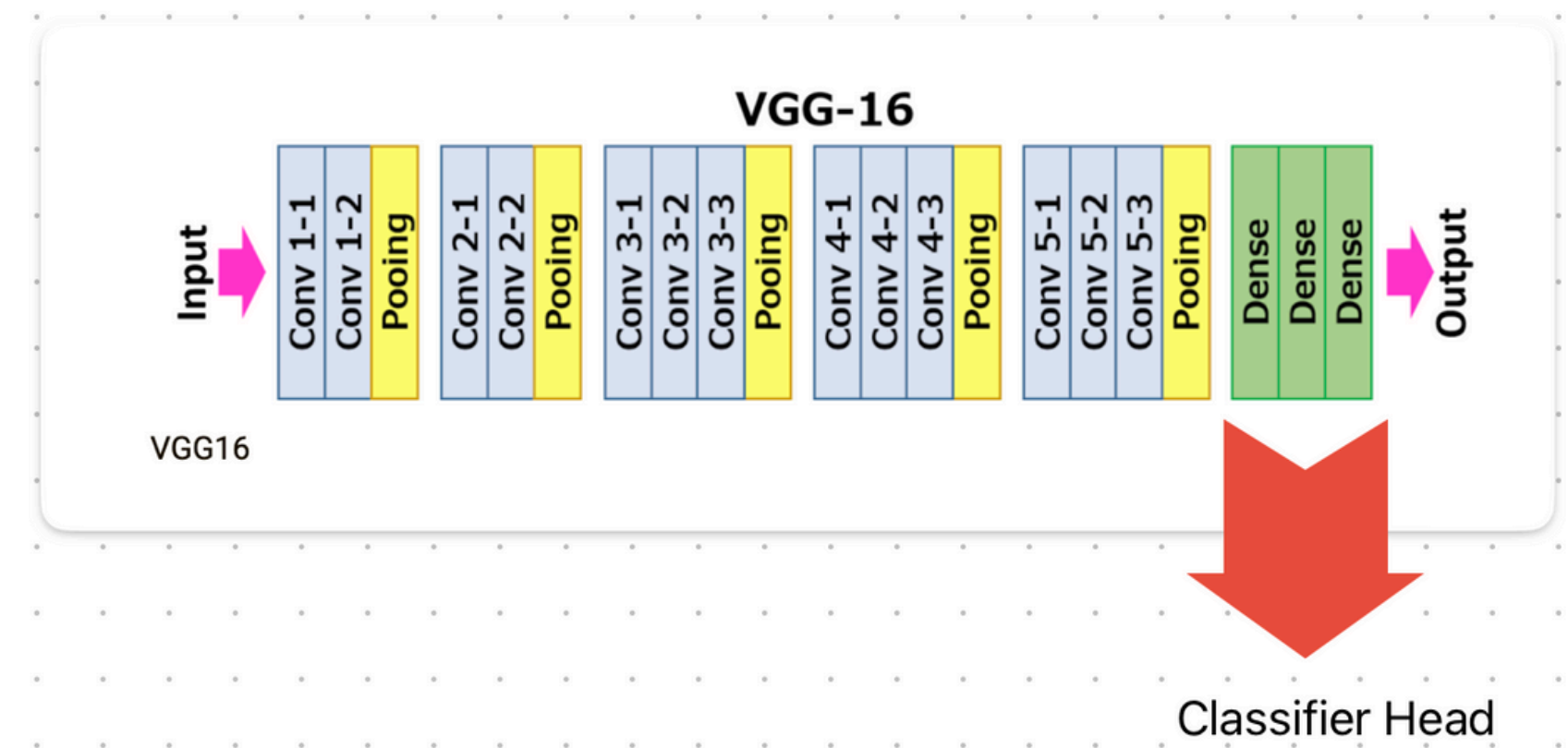
Preprocessing:

- Normalized data.
- Converted categorical labels to numerical values.
- Prepared as NumPy arrays.



MODEL 1 ARCHITECTURE-VGG-16

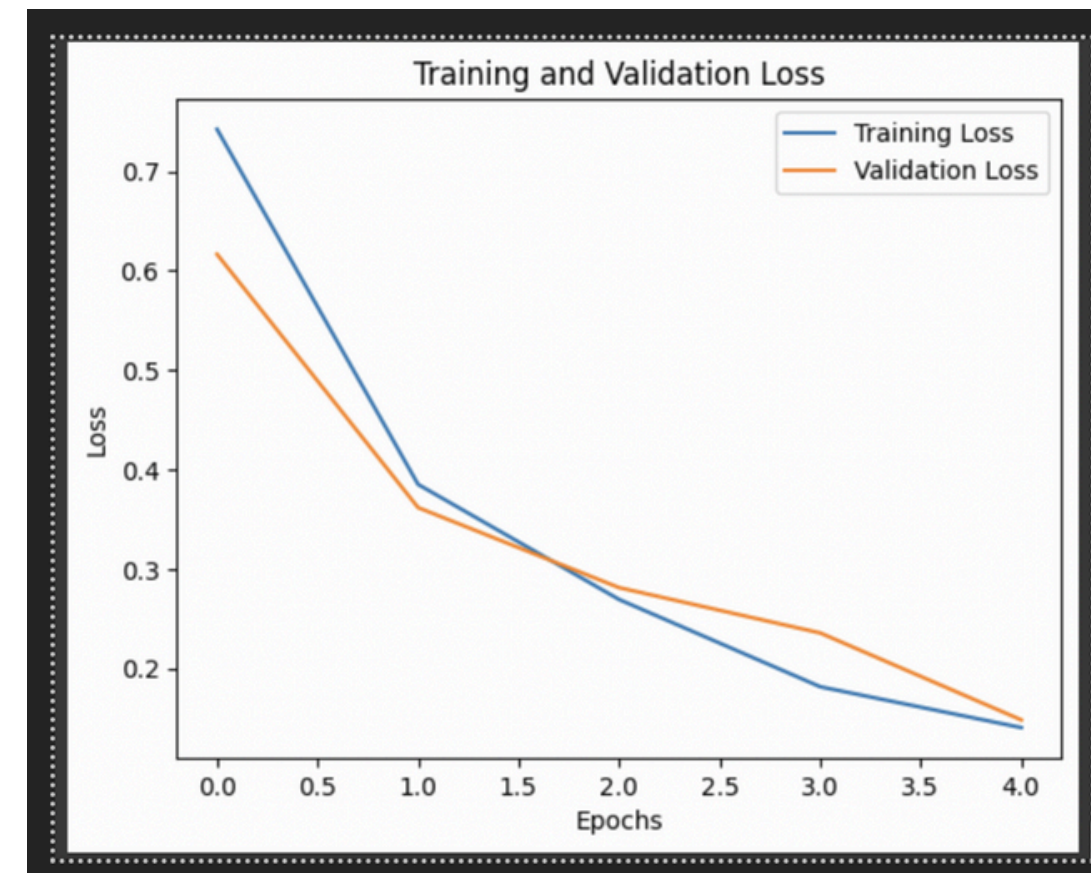
- **Base Model:** Pre-trained VGG16 (used for feature extraction).
- **Frozen Layers:** Convolutional layers frozen for feature extractor
- **Modified Classifier Head:**
 - Fully Connected Layer (Input: 25,088, Output: 256)
 - Activation: ReLU
 - Dropout rate: 0.5
 - Fully Connected Layer (Input: 256, Output: 3)
 - Output: 3-class classification.
 - Total parameters: 21,138,243
 - Trainable Parameters: 6,423,555



MODEL 1-VGG16 RESULTS

- During training we experimented with different optimizers, learning rates, batch sizes, and epochs.
- Model is finally retrained with best parameters: batch size: 32, learning rate: 0.0001, epoch: 5, optimizer: Adam
- Achieved **98% accuracy** on the test set with strong precision, recall, and F1-scores across all classes.
- Class 2 had perfect metrics (**1.00**) while Class 0 and Class 1 showed slight variability with metrics around **0.97-0.98**
- Training and validation loss decreased steadily with minimal gap, indicating good generalization and no clear signs of overfitting.
- Final loss convergence reflects stable and effective training.

Classification Report:				
	precision	recall	f1-score	support
Class 0	0.98	0.98	0.98	125
Class 1	0.97	0.97	0.97	60
Class 2	1.00	1.00	1.00	65
accuracy			0.98	250
macro avg	0.98	0.98	0.98	250
weighted avg	0.98	0.98	0.98	250

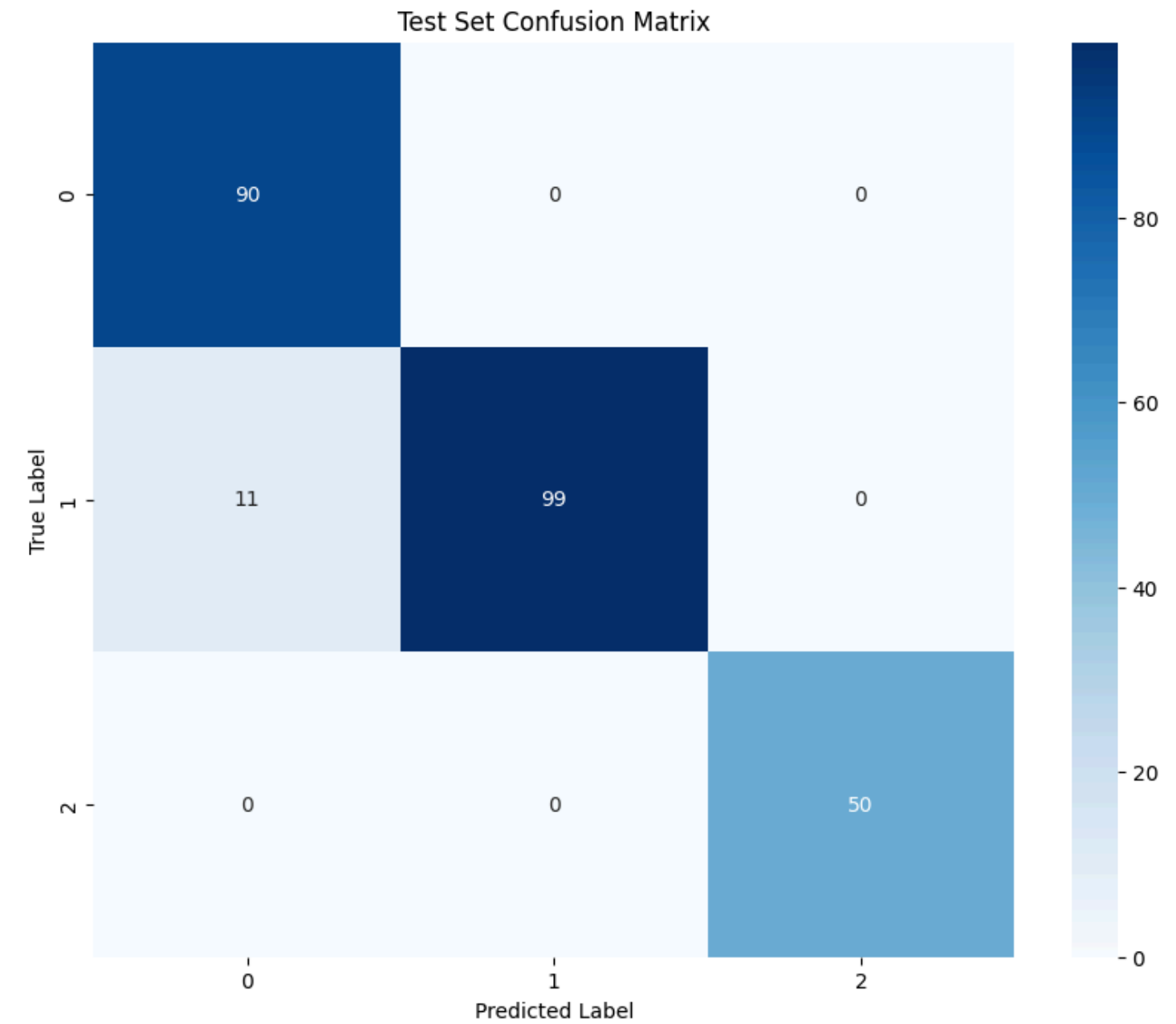
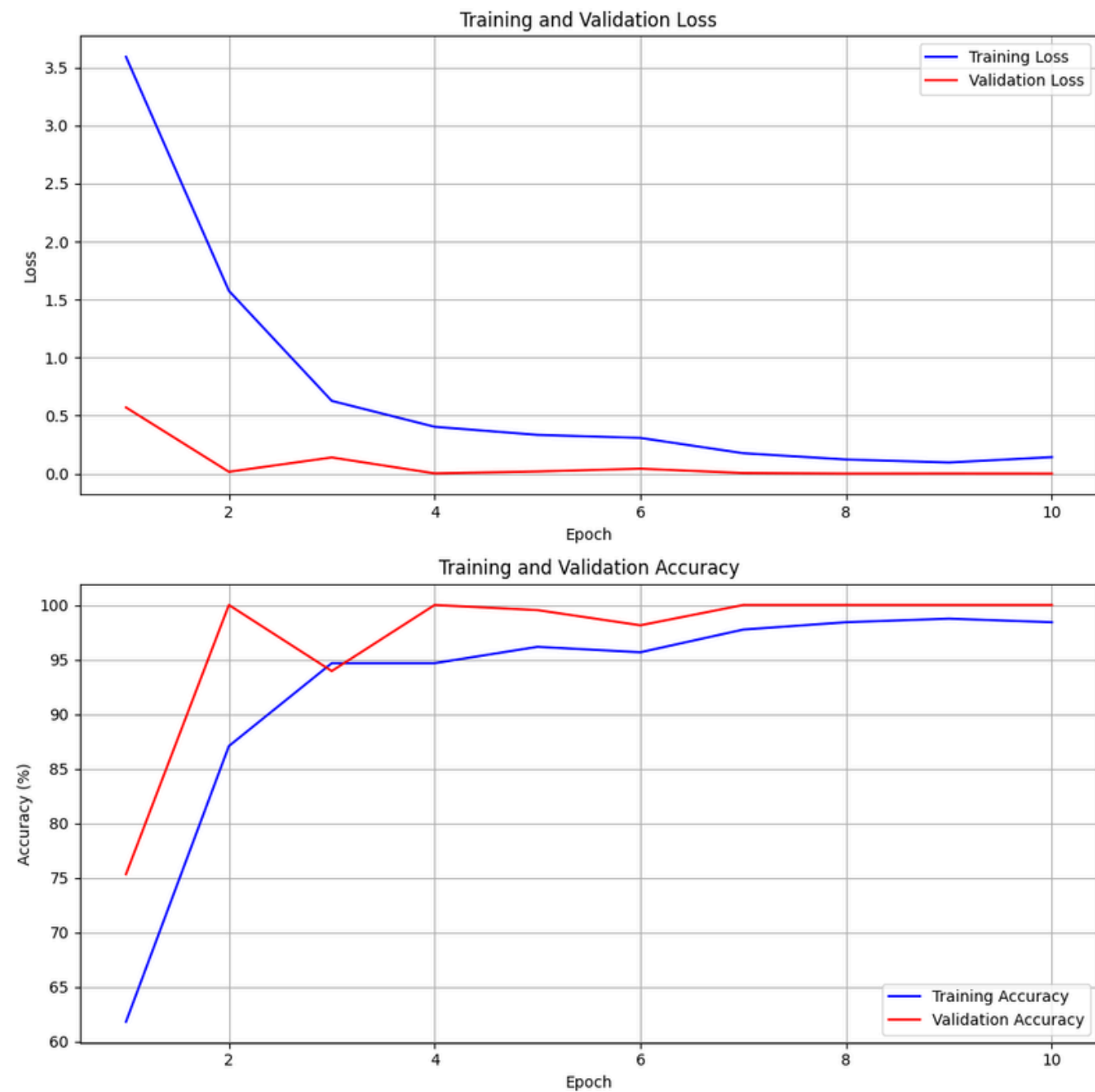


INCEPTION V3

Property	Value
Model Type	Inception V3 (pretrained)
Input Size	299 x 299 x 3
Total Layers	298
Training Strategy	Fine-tuning last 2 inception blocks
Layer Type	Specifications
Batch Normalization	Features: 2048
Dropout	Rate: 0.4
Linear	2048 → 512
ReLU	Activation
Batch Normalization	Features: 512
Dropout	Rate: 0.4
Linear	512 → 3

Block	Parameters	Layer Count
FROZEN LAYERS		
Conv2d_1a_3x3	928	3
Conv2d_2a_3x3	9,280	3
Conv2d_2b_3x3	18,560	3
Conv2d_3b_1x1	5,280	3
Conv2d_4a_3x3	138,624	3
Mixed_5b	255,904	21
Mixed_5c	277,472	21
Mixed_5d	285,152	21
AuxLogits	3,326,696	8
TRAINABLE LAYERS		
Mixed_6a	1,153,280	12
Mixed_6b	1,297,408	30
Mixed_6c	1,691,008	30
Mixed_6d	1,691,008	30
Mixed_6e	2,141,952	30
Mixed_7a	1,698,304	18
Mixed_7b	5,044,608	27
Mixed_7c	6,076,800	27
fc	1,055,747	8
Parameter Type	Count	Percentage
Total Parameters	26,168,011	100%
Trainable Parameters	21,850,115	83.5%
Frozen Parameters	4,317,896	16.5%

MODEL 2-INCEPTION V3 RESULTS



Model 3 Architecture-ResNET-50

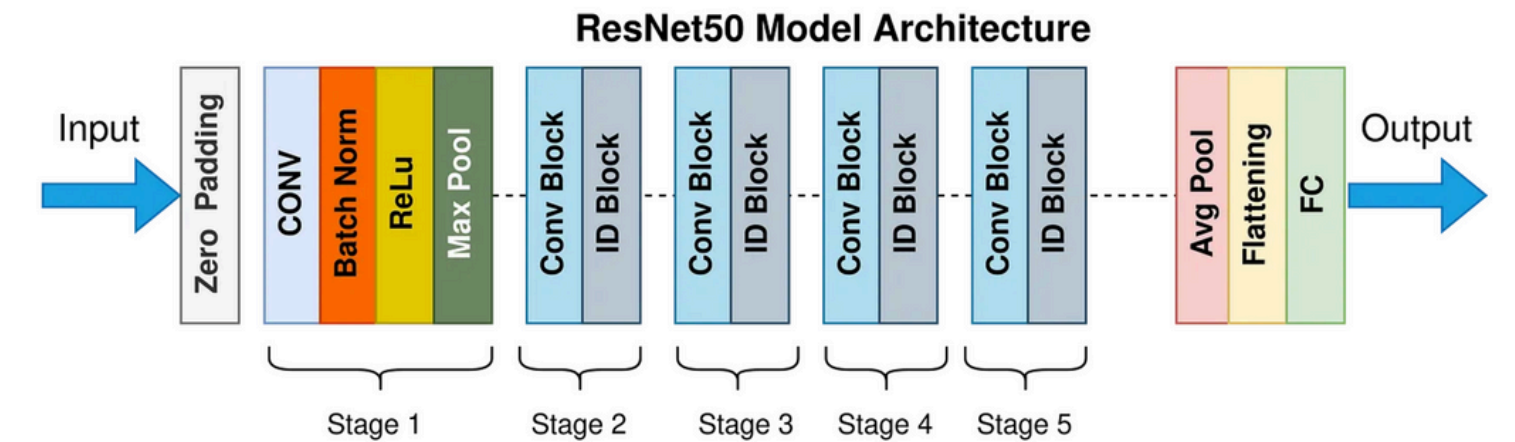
Base Model: Used ResNet-50 pretrained on ImageNet as the base model

Layer Freezing : Froze all layers except the final residual block (layer4), enabling fine-tuning of higher-level features while preserving pretrained weights in earlier layers.

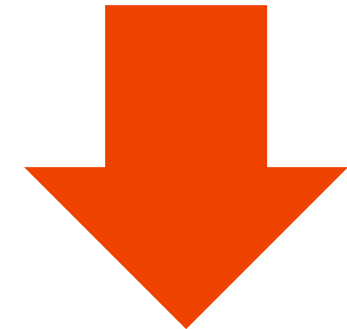
Modified Classifier Head:

Replaced the default fully connected (FC) layer with a new classifier:

1. Batch Normalization (nn.BatchNorm1d)
 - Stabilizes learning and speeds up convergence.
2. Intermediate Fully Connected Layer (nn.Linear)
 - Input: Features from ResNet-50 (e.g., num_features).
 - Output: 512 neurons.
 - Activation: ReLU for introducing non-linearity.
3. Final Fully Connected Layer (nn.Linear)
 - Input: 512 neurons.
 - Output: 3 classes
4. Dropout (nn.Dropout): Applied with a rate of 0.5 to reduce overfitting.



Resnet-50 Model architecture



Classified Head

ResNET-50 Results

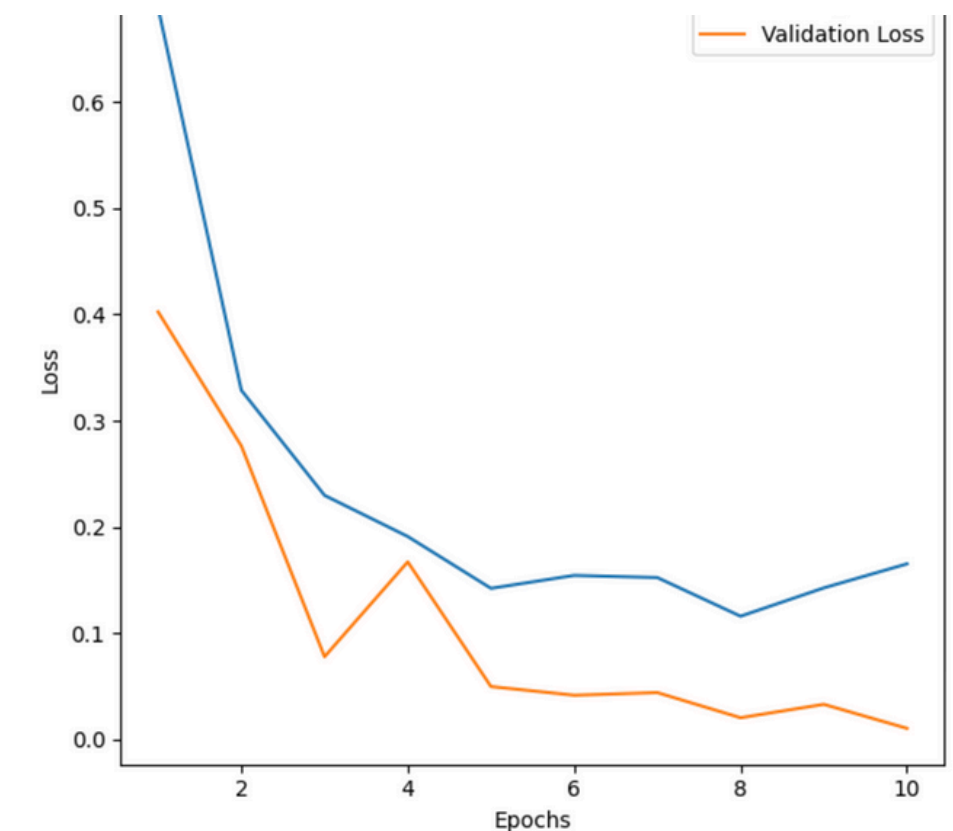
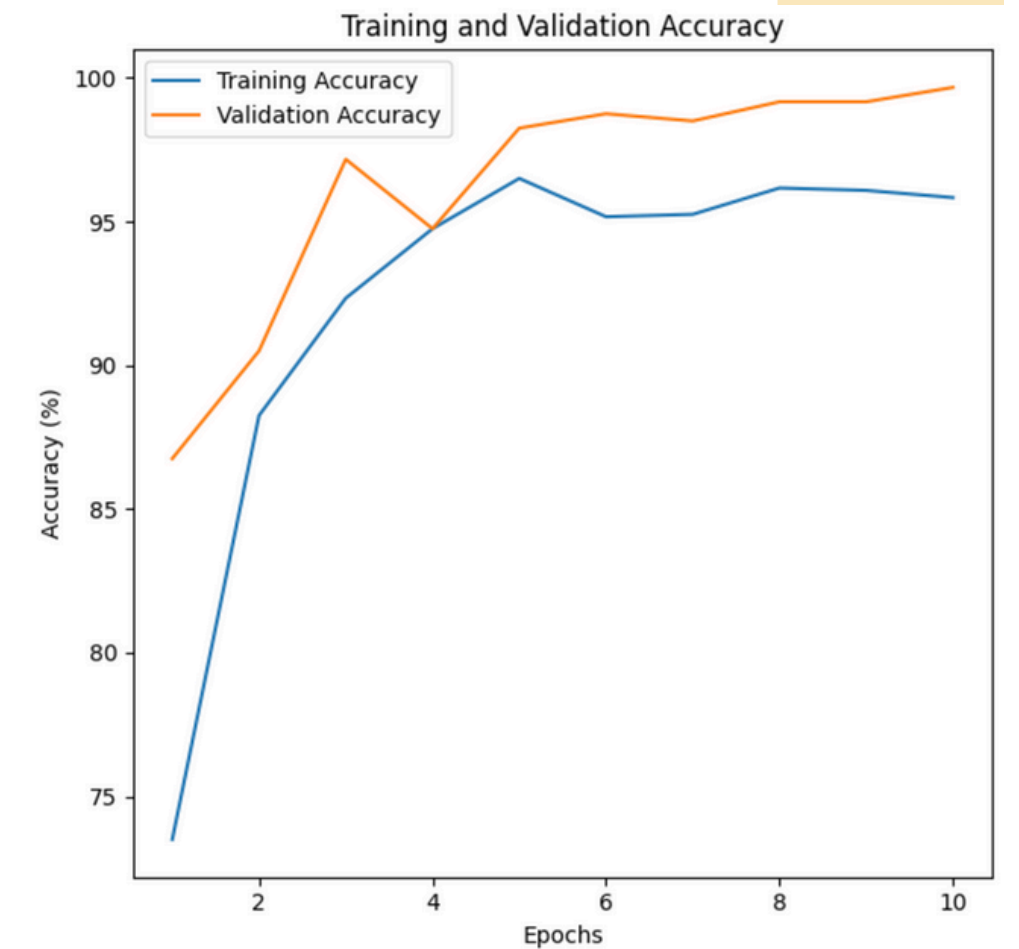
Model: ResNet-50 deep learning model trained over 10 epochs.

Dataset: Split into training and validation sets, with metrics tracked for both.

Learning Rate: Constant at 0.001 throughout training.

Metrics Observed:

- Train Loss: Steadily decreased, indicating effective learning.
- Validation Loss: Consistently decreased, showing good generalization.
- Train Accuracy: Improved from 73.5% to 95.83%.
- Validation Accuracy: Increased from 86.75% to 99.67%, demonstrating excellent performance on unseen data.



Model Comparison: ResNet-50, VGG16, & INCEPTION V3

Metric	ResNet-50	VGG16	Inception V3
Training Accuracy	95.83%	96.97%	98.75%
Validation Accuracy	99.67%	96.74%	100.00%
Training Loss	0.1652	0.1405	0.0958
Validation Loss	0.0104	0.1482	0.0002
Testing Aucuracy	34 %	98.4 %	95. 6%



Conclusion and Future Work

Conclusion:

- Among the three models, VGG16 and Inception V3 delivered strong performance, achieving high accuracy and robust metrics. Despite comparable parameter sizes across all models, VGG16 stood out with fewer trainable parameters and faster training times, yet still slightly outperformed Inception V3. In contrast, ResNet50 required longer training and showed limited success

Future Work

- Experimenting with other image types like MRIs
- Fine tuning more layers for some of our models to achieve an even better accuracy
- Increase the amount of training and testing data